Derivatives

1. Motivation

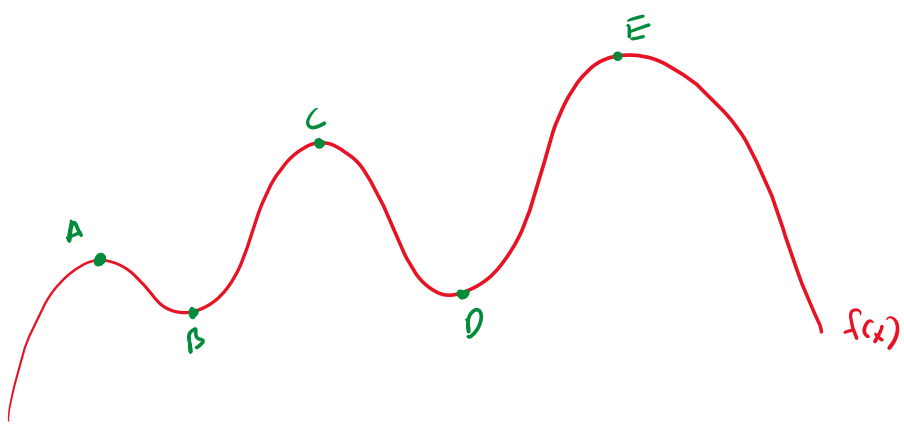
If $f(x)$ is a quadratic then the graph is

and we can find the maximum or minimum of $f(x)$.

what if $f(x)$ is not a quadratic ?

How can we find the max / min of $f(x)$?

we use " Derivatives" to do that.

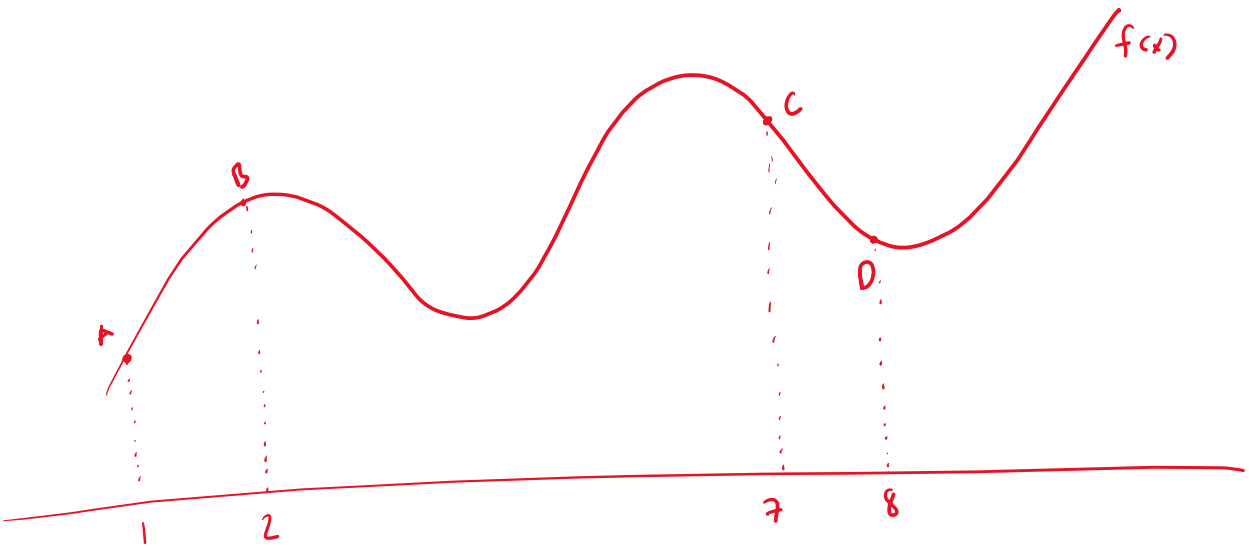


using " Derivatives" we can find the locations of A, B, C, D and E . Hence, we can find the max of $f(x)$.

2. The idea of “derivative”

The derivative of a function f(x) is denoted by f’(x) [reads: f prime of x]. And f’(a) measures the instant change of f(x) at the point a.

- If f’(a) >0 then f(x) is increasing at x = a.
- If f’(a) <0 then f(x) is decreasing at x = a.
- The absolute value of f’(a) is the speed of increasing or decreasing at x = a.



$f'(1) > 0$ (f(x) going up at x = 1)

$f'(2) > 0$

$f'(7) < 0$, $f'(8) < 0$

$f'(1) > f'(2)$ (f(x) going up faster at x = 1)

$|f'(7)| > |f'(8)|$ (f(x) going down faster at x = 7)

3. Calculating derivatives

We will use several "rules" to calculate the derivative of a function.

<u>Rule 1</u> : Constant Rule

If $f(x) = c$ (constant)

$\Rightarrow f'(x) = 0$

<u>OR</u> : $(c)' = 0$

<u>Rule 2</u> : Linear Rule

If $f(x) = c \cdot x$ (linear function)

$\Rightarrow f'(x) = c$

<u>OR</u> : $(cx)' = c$

Example : $(4x)' = 4$

$(2026x)' = 2026$

<u>Rule 3</u> : Power Rule

If $f(x) = x^n$ (power function)

$\Rightarrow f'(x) = n \cdot x^{n-1}$

<u>OR</u> : $(x^n)' = n \cdot x^{n-1}$

Example : Find $f'(x)$

① $f(x) = x^{10} \Rightarrow f'(x) = 10 \cdot x^{10-1}$

$\Rightarrow f'(x) = 10 \cdot x^{9}$

② $f(x) = x^{3} \Rightarrow f'(x) = 3x^{3-1}$

$f'(x) = 3x^{2}$

③ $f(x) = x^{2} \Rightarrow f'(x) = 2x^{2-1} = 2x$

④ $f(x) = \frac{1}{x^{7}} = x^{-7}$

$$\boxed{\text{Formula : } \frac{1}{x^{k}} = x^{-k}}$$

$\Rightarrow f'(x) = -7 \cdot x^{-7-1}$

$\Rightarrow f'(x) = -7x^{-8} = \frac{-7}{x^{8}}$

⑤ $f(x) = \frac{1}{x} = x^{-1}$

$\Rightarrow f'(x) = -1 \cdot x^{-1-1} = -x^{-2}$

$= \frac{-1}{x^{2}}$

Practice: Find $f'(x)$

① $f(x) = 2026$

② $f(x) = -7x$

③ $f(x) = \frac{1}{3}x$

④ $f(x) = x^{100}$

⑤ $f(x) = x^{20}$

⑥ $f(x) = \frac{1}{x^9}$

⑦ $f(x) = \frac{1}{x^{5/3}}$

Rule 4: Scalar Multiplication

$$[K \cdot f(x)]' = K \cdot f'(x)$$

Example:

① $f(x) = 8 \cdot x^3$

$\Rightarrow f'(x) = (8x^3)'$

$= 8 \cdot (x^3)'$

$= 8 \cdot 3 \cdot x^{3-1}$

$= 24x^2$

②

$f(x) = \frac{1}{3}x^6$

$f'(x) = \left(\frac{1}{3} \cdot x^6\right)'$

$= \frac{1}{3}(x^6)'$

$= \frac{1}{3} \cdot 6x^{6-1}$

$= 2x^5$

③ $f(x) = \frac{x^{10}}{7}$

$$f'(x) = \left(\frac{x^{10}}{7}\right)' = \frac{(x^{10})'}{7} = \frac{10x^9}{7}$$

④ $f(x) = \frac{1}{3x^4}$

$$f'(x) = \left(\frac{1}{3x^4}\right)'$$

$$= \left(\frac{1}{3} \cdot \frac{1}{x^4}\right)' = \frac{1}{3}\left(\frac{1}{x^4}\right)'$$

$$= \frac{1}{3}\left(x^{-4}\right)'$$

$$= \frac{1}{3} \cdot (-4) \cdot x^{-4-1}$$

$$= -\frac{4}{3} \cdot x^{-5} = -\frac{4}{3x^5}$$

<u>Rule 5</u> : Addition Rule

$$[f(x) + g(x)]' = f'(x) + g'(x)$$

$$[f(x) - g(x)]' = f'(x) - g'(x)$$

Example :

① $f(x) = x^{10} + 6x^4 + 3x - 7$

$$f'(x) = \left[x^{10} + 6x^4 + 3x - 7\right]'$$

$$= (x^{10})' + (6x^4)' + (3x)' - (7)'$$

$$= 10x^9 + 24x^3 + 3$$

② $$f(x) = \frac{x^{10} + 3x^7 + 1}{x^4}$$

$$f(x) = \frac{x^{10}}{x^4} + \frac{3x^7}{x^4} + \frac{1}{x^4}$$

$$f(x) = x^{10-4} + 3x^{7-4} + x^{-4}$$

$$\boxed{\text{Formula : } \frac{x^m}{x^n} = x^{m-n}}$$

$$f(x) = x^6 + 3x^3 + x^{-4}$$

$$f'(x) = \left(x^6 + 3x^3 + x^{-4}\right)'$$

$$= 6x^{6-1} + 3 \cdot 3x^{3-1} - 4x^{-4-1}$$

$$= 6x^5 + 9x^2 - \frac{4}{x^5}$$

$$= 6x^5 + 9x^2 - 4x^{-5}$$

<u>Practice</u>: Find $f'(x)$

① $f(x) = 6x^4$

② $f(x) = 9x^2$

③ $f(x) = \frac{1}{3}x^6$

④ $f(x) = 4x^3 + 2x^2 - 9x + 1$

⑤ $f(x) = \dfrac{x^3 + 2x^2 + 1}{x^2}$

<u>Rule 6</u>: Quotient Rule.

$$\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$$

<u>Example</u>:

$$h(x) = \frac{x^3 + 2x + 1}{x^2}$$

$$h'(x) = \frac{(x^3+2x+1)' \cdot x^2 - (x^3+2x+1) \cdot (x^2)'}{(x^2)^2}$$

$$= \frac{(3x^2+2) \cdot x^2 - (x^3+2x+1) \cdot 2x}{x^4}$$

<u>practice</u> : Find $f'(x)$

$$f(x) = \frac{x^3+2x^2}{x^6+9}$$

$$\left(\frac{\text{Top}}{\text{Bot}}\right)' = \frac{(\text{Top})' \cdot \text{Bot} - \text{Top} \cdot (\text{Bot})'}{(\text{Bot})^2}$$

<u>Rule 7</u> : Product Rule

$$(\text{First} * \text{Second})' = (\text{First})' \cdot \text{Second} + \text{First} \cdot (\text{Second})'$$

<u>Example</u> : Find $f'(x)$

$$f(x) = (x^3 + x^2 + 2) \cdot (2x^4 + x^9 - 6x)$$

$$\Rightarrow f'(x) = (x^3 + x^2 + 2)' \cdot (2x^4 + x^9 - 6x) + (x^3 + x^2 + 2) \cdot (2x^4 + x^9 - 6x)'$$

$$= (3x^2 + 2x) \cdot (2x^4 + x^9 - 6x) + (x^3 + x^2 + 2) \cdot (8x^3 + 9x^8 - 6)$$

<u>Practice</u>: Find $f'(x)$

① $f(x) = (x^3 + 2x) \cdot (6x + 9)$

② $f(x) = (x^{10} - 9) \cdot (x^{11} + 3x)$